

Coding & Solution

Date	15 March 2026
Team ID	xxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

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This section evaluates the quality and completeness of the implemented code and the overall solution, demonstrating working functionality, clean code practices, and alignment with the defined requirements.

Solution Summary

Field	Details
Repository Link / URL	[Add your GitHub / GitLab repository link here]
Programming Language(s)	Python, JavaScript
Framework(s) Used	Streamlit, OpenAI Whisper, Sentence-Transformers (Sentence-BERT), Librosa, Web Speech API (HTML/JS version)
Key Features Implemented	Audio upload and speech-to-text transcription using OpenAI Whisper Semantic similarity scoring between transcription and reference concept using Sentence-BERT Filler word, pause ratio, and RMS energy analysis using Librosa Conceptual understanding score with qualitative feedback (Strong / Moderate / Poor) Interactive Streamlit dashboard with waveform visualization Downloadable PDF report generation with metrics and summaries Self-contained HTML/JS web version using the Web Speech API
Pending / Incomplete Features	Multi-language support and real-time streaming transcription are planned for future enhancement
Setup / Run Instructions	1. Clone the repository and navigate into the project folder 2. Install dependencies: <code>pip install -r requirements.txt</code> 3. Run the application: <code>streamlit run app.py</code> 4. Open the local URL shown in the terminal (default <code>http://localhost:8501</code>) 5. For the standalone web version, open the HTML file directly in a Chrome-based browser

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application integrates multiple AI and audio-processing modules (Whisper, Sentence-BERT, Librosa) into a single Streamlit-based educational assessment tool, with a lightweight HTML/JS companion version for quick browser-based use.